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ANALYSIS OF 3D VEL GRAPHS USING THE X5 VARIABLE AND WITH NEGATIVE VALUES OF OTHER VARIABLES

Two years ago, the Nobel Prize in Economics was awarded to three American economists who significantly improved the understanding of the role of banks in the economy, especially during financial crises, by giving money to banks and the population to restore purchasing activity in the country. Here it is worth recalling at once that in a number of countries, such as Sweden, Switzerland, Denmark and Japan, banks issued some customers with a negative rate, i.e. customers returned an amount less than they received [1, 2, 3].

This article considers the issue of calculating the values of the lower critical volume of the Vel spherical economic shell and constructing 3D drawings when the values of three variables change, i.e. Vel (GDPel) = f(X1, X2, X3, X4, X5). In some cases, when the Vel values had small minima and maxima compared to the last calculated Vel value, then 3D drawings were built from the calculation so that the reader would understand more clearly how these changes occur.

Here, Figure 1 shows a 3D graph of Vel, when the values of the variables were as follows X1= X3= -1...-10, X2= 1, X4= 1...10, X5= 1.36...14.14. Here the values of Vel increase by 1083.17 times. In the following Figure 2 the depicted 3D Vel graph with variables X1= X2= 1, X3= -1...-0.1, X4= 10...1, X5= 14.15...1.88 decreases by 563.82 times.

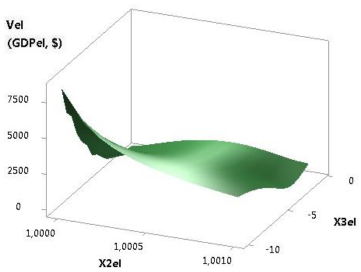

Fig. 1. 3D of Vel (GDPel)

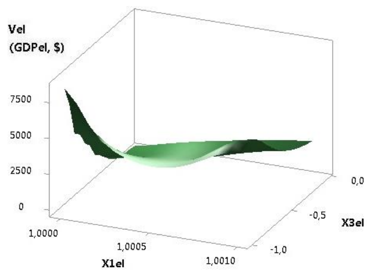

Fig. 2. 3D of Vel (GDPel)

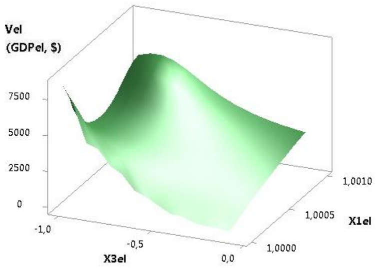

Fig. 3. 3D of Vel (GDPel)

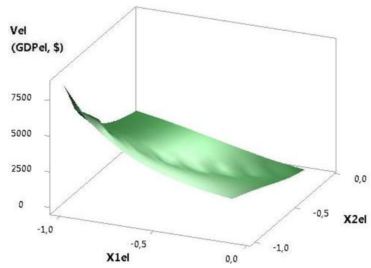

Fig. 4. 3D of Vel (GDPel)

The following two figures 3 and 4 show 3D Vel graphs when the variables were X1= 1, X2= X3=-1...-0.1, X4= 10.1, X5= 14.1.42 and X1= X2= X3= -1...-0.1, X4= 10.1, X5= 14.14.1.41. As we can see, the plotted 3D Vel graph in Fig. 3 decreases significantly by 993.08 times, and in Fig. 4 by 1000 times.

The constructed 3D Vel graph in Figure 5 decreases by 1083.17 times at variables X1= X3=-1...-0.1, X2= 1, X4= 10...1, X5= 14.14...1.36. From the following Figure 6, it can be seen that at variables X1= X3= 1, X2= -1...-0.1, X4= 10...1, X5= 14.14...1.41, the Vel graph decreases by 1000 times.

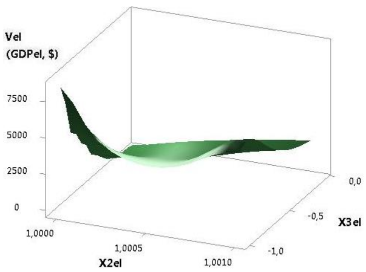

Fig. 5. 3D of Vel (GDPel)

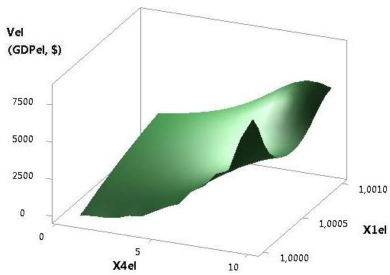

Fig. 6. 3D of Vel (GDPel)

Figures 7 and 8 were plotted at X1= -1...-0.1, X2= X3= 1, X4= 10...1, X5= 14.14...1.42 and X1= X3=-1...-10, X2= 1, X4= 10...1, X5=14.14...1.36. Here in Fig. 7 the Vel value decreases by 993.08 times, and in Figure 8 it decreases by 1083.17 times.

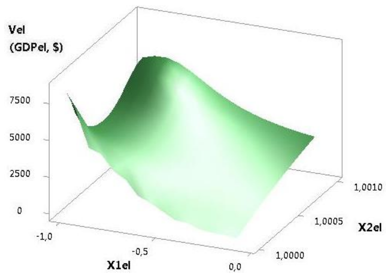

Fig. 7. 3D of Vel (GDPel)

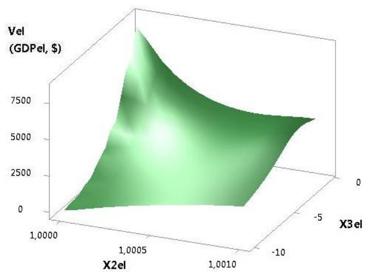

Fig. 8. 3D of Vel (GDPel)

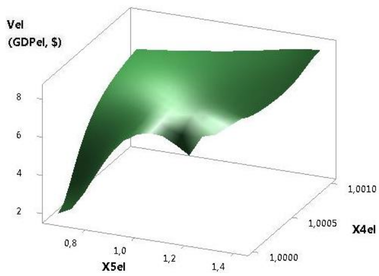

Fig. 9. 3D of Vel (GDPel)

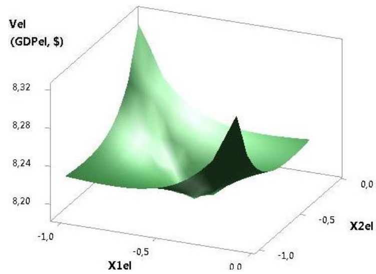

Fig. 10. 3D of Vel (GDPel)

The following Figure 9 was plotted at X1= X2= -0.1...-1, X3= -1...-0.1, X4= 1, X5=1.41...0.67 decreases slightly by 4.44 times. Figure 10 plotted at X1= -0.1...-1, X2= X3= -1...-0.1, X4= 1, X5=1.41 has minima of 8.19 at points 5 and 6.

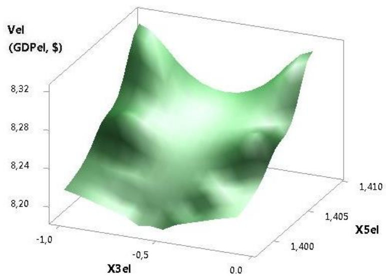

Fig. 11. 3D of Vel (GDPel)

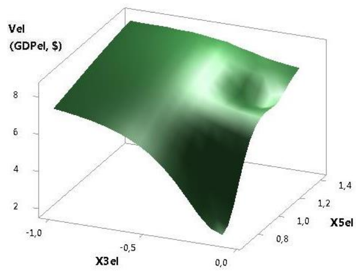

Fig. 12. 3D of Vel (GDPel)

The last figures 11 and 12 were plotted at X1= -1...-0.1, X2= X3= -0.1...-1, X4=0.71...0.71, X5=-1 and X1= X2= -1...-0.1, X3= -0.1...-1, X4=1, X5=0.67...1.41. In these examples, Figure 11 the Vel variable has minima of 8.19 at points 5 and 6, and in Figure 12 it increases slightly by 4.44 times.

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